Configuração 1: M = 128, C = 8, K = 8, IoT = 10 dB, 16-QAM
Perda de SNR do BCD-MMSE (4 iterações) em SER = 10^{-2}, em dB, em relação à referência ideal (float64); × = não chega a 10^{-2} nas SNRs simuladas
Fundo laranja no rótulo: o modo com o melhor resultado no par, comparando do formato com menos bits para o de mais, sem contar as colunas em que todos os modos ficam abaixo de 0,05 dB
Embaixo da perda: maior ocupação da XCU50 entre LUT, DSP, FF e BRAM (síntese HLS, sem diretivas), no mesmo formato e modo

Perda de SNR (dB)

< 0,1 dB | 0,1 a 0,5 dB | 0,5 a 1 dB | 1 a 2 dB | 2 a 3 dB | 3 a 5 dB | 5 a 10 dB | > 10 dB | × não chega a 10^{-2} | rótulo do melhor modo

TDLA30-5

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	0.05 1.8%	0.00 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	0.05 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.05 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.05 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.05 1.8%	0.00 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.01 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.03 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.03 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.03 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.03 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.03 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.03 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.03 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.03 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.03 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.03 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.03 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.03 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.03 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.03 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.03 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.02 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.02 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.02 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.02 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.02 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.02 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.02 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-75

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	0.06 1.8%	0.00 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	0.06 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.06 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.06 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.06 1.8%	0.00 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	0.00 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.00 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.00 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.00 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.00 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.01 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.01 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.01 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.01 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLA30-300

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	0.05 1.8%	0.00 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	0.05 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.05 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.05 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.05 1.8%	0.00 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	0.00 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.00 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.00 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.00 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.00 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.00 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.00 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.00 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.00 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.00 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.01 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.01 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.00 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.00 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.00 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.00 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.00 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.00 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.00 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.00 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.00 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.00 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLC300-100

Quantização	Estouro	<24,12>	<28,12>	<30,14>	<34,14>
AP_TRN	AP_SAT	0.05 1.8%	0.00 2.1%	0.00 3.1%	0.00 3.6%
	AP_SAT_ZERO	0.05 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.05 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.05 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.05 1.8%	0.00 2.2%	0.00 3.7%	0.00 4.2%
AP_TRN_ZERO	AP_SAT	0.01 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.01 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.01 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.01 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.01 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND	AP_SAT	0.02 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.02 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.02 1.7%	0.00 2.3%	0.00 3.5%	0.00 4.1%
	AP_WRAP_SM	0.02 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_ZERO	AP_SAT	0.02 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.02 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.02 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.02 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_MIN_INF	AP_SAT	0.02 1.9%	0.00 2.2%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.02 2.1%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.02 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
AP_RND_INF	AP_SAT	0.02 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.02 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.02 1.6%	0.00 2.0%	0.00 3.2%	0.00 3.7%
	AP_WRAP_SM	0.02 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%
AP_RND_CONV	AP_SAT	0.02 2.0%	0.00 2.3%	0.00 3.7%	0.00 4.2%
	AP_SAT_ZERO	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_SAT_SYM	0.02 2.2%	0.00 2.5%	0.00 3.7%	0.00 4.2%
	AP_WRAP	0.02 1.8%	0.00 2.4%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.02 2.0%	0.00 2.4%	0.00 3.7%	0.00 4.2%

TDLB100-400

Quantização	Estouro	<22,10>	<26,10>	<32,12>	<36,12>
AP_TRN	AP_SAT	0.00 1.7%	0.00 2.0%	0.00 3.3%	0.00 3.7%
	AP_SAT_ZERO	0.00 1.6%	0.00 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	0.00 1.8%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP	0.00 1.6%	0.00 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.00 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
AP_TRN_ZERO	AP_SAT	0.47 1.9%	0.03 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.47 1.7%	0.03 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.47 2.1%	0.03 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.47 1.4%	0.03 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	0.47 1.9%	0.03 2.3%	0.00 3.9%	0.00 4.4%
AP_RND	AP_SAT	0.02 1.8%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.02 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.02 2.0%	0.00 2.4%	0.00 3.9%	0.00 4.4%
	AP_WRAP	0.02 1.6%	0.00 2.1%	0.00 3.7%	0.00 4.2%
	AP_WRAP_SM	0.02 1.9%	0.00 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_ZERO	AP_SAT	0.02 1.9%	0.00 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.02 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.02 2.1%	0.00 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.02 1.4%	0.00 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	0.02 1.9%	0.00 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_MIN_INF	AP_SAT	0.02 1.8%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.02 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.2%
	AP_SAT_SYM	0.02 2.0%	0.00 2.4%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.02 1.6%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	0.02 1.9%	0.00 2.2%	0.00 3.9%	0.00 4.4%
AP_RND_INF	AP_SAT	0.02 1.9%	0.00 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.02 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.02 2.1%	0.00 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.02 1.4%	0.00 1.9%	0.00 3.4%	0.00 3.9%
	AP_WRAP_SM	0.02 1.9%	0.00 2.3%	0.00 3.9%	0.00 4.4%
AP_RND_CONV	AP_SAT	0.02 1.9%	0.00 2.3%	0.00 3.9%	0.00 4.4%
	AP_SAT_ZERO	0.02 1.7%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_SAT_SYM	0.02 2.1%	0.00 2.5%	0.00 3.9%	0.00 4.3%
	AP_WRAP	0.02 1.6%	0.00 2.2%	0.00 3.9%	0.00 4.4%
	AP_WRAP_SM	0.02 1.9%	0.00 2.2%	0.00 3.9%	0.00 4.4%